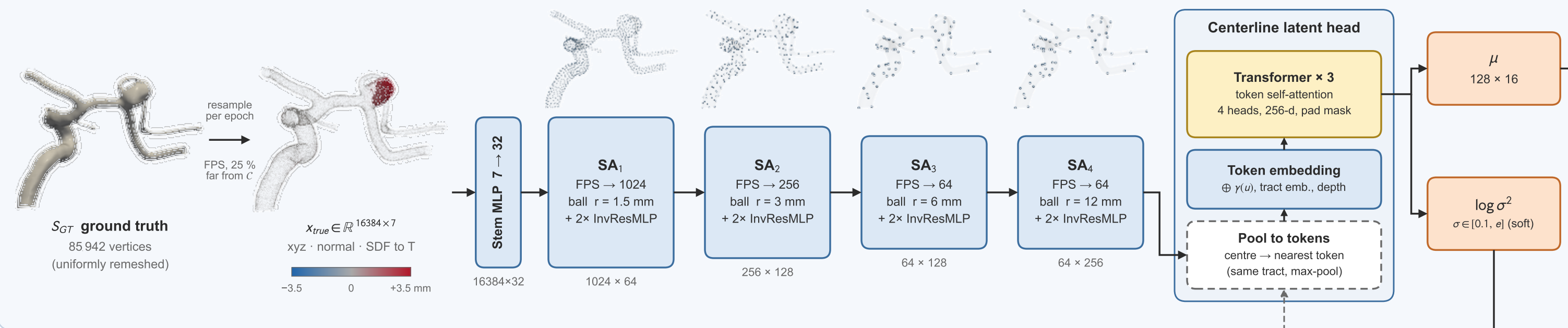


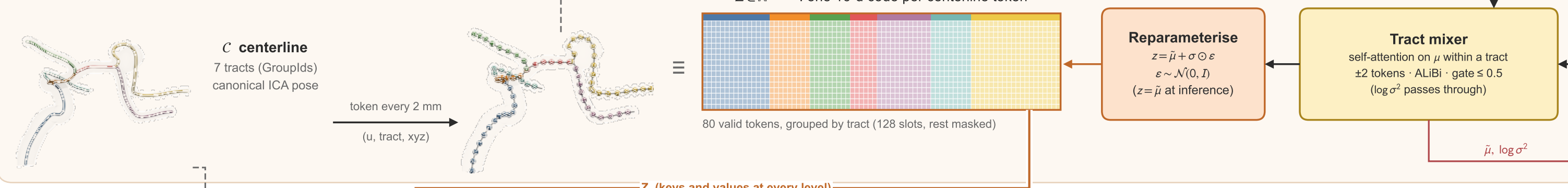
Stage-2 geometry VAE: training pipeline

Hierarchical PointNeXt encoder → tree-valued centerline latent Z → progressive SplineConv decoder that deforms the template mesh. Renders: case p462 from cleandata, canonical pose, mm.

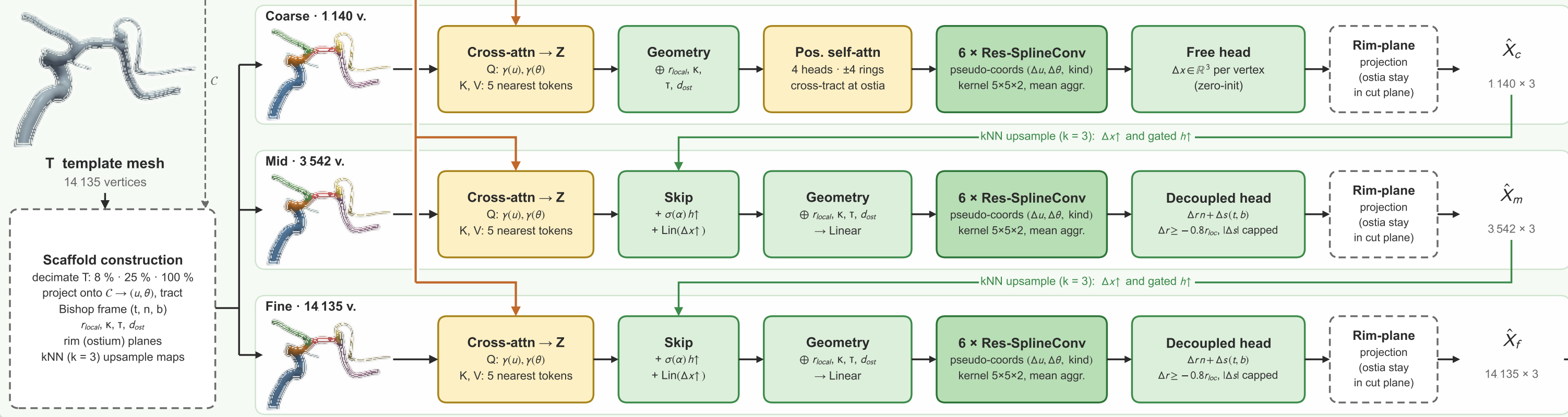
(a) Encoder $q_\phi(Z|S_{GT}, C)$



(b) Tree-valued latent



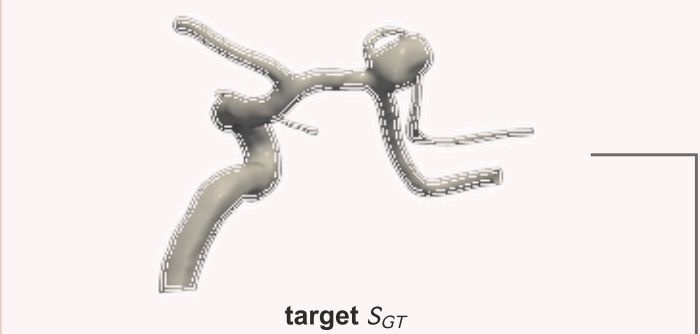
(c) Decoder $p_\theta(\hat{S}|Z, T, C)$



(d) Training objective

$$\mathcal{L} = \mathcal{L}_{CD} + \mathcal{L}_{rad} + \lambda_{KL} \beta \mathcal{L}_{KL} + 0.15 \mathcal{L}_{disp} + 0.05 \mathcal{L}_{lap} + 0.02 \mathcal{L}_n$$

AdamW $2 \cdot 10^{-4}$, cosine · EMA 0.993
batch 4×8 accum. · 200 epochs



Legend

- encoder layer (learned)
- attention block
- latent / stochastic
- decoder layer (learned)
- spline convolution
- fixed geometric operation
- loss term
- latent Z broadcast

$\mathcal{L}_{KL}$ per-token KL to $\mathcal{N}(0, I)$
mean over valid tokens
 β by GECO, target 12 nats / token

Regularisers on the mesh
 $\mathcal{L}_{disp}$ Dirichlet energy of ($\Delta r, \Delta s$)
 $\mathcal{L}_{lap}$ Laplacian smoothing
 $\mathcal{L}_n$ normal consistency
(fold, stretch: logged, weight 0)

$\mathcal{L}_{rad}$ radial Huber
composed radius vs ray-cast r^*
fine + 0.5 · mid

$\mathcal{L}_{CD}$ weighted Chamfer
Huber point-to-plane + 0.2 · L2
fine + 0.5 · mid + 0.05 · coarse
weights from distance to C (cap 4)
16 384 area samples of $\hat{X}$